

The closed-walk law for the Fermat block-rank factor

An identification of the Degtyarev–Shimada rank polynomial as the constant term of a closed lattice walk, with the parity correction realized as a rest step, proved byte-exact for both parities

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Companion to [THE_LOCALIZATION_THEOREM.md](#) and [THE_WITHIN_PAIR_FUNCTOR.md](#). Localization proves *where* any torsion of a Fermat cell must live (one CRT block); the within-pair functor opens the *interior* of one block; this note names the *value* every block contributes. It identifies the Degtyarev–Shimada rank polynomial — the combinatorial object that governs the per-block survivor count on the author’s computational campaign — as the constant term of a closed lattice walk, settling the coupling mechanism (“the meshing”) that was a measured pattern across the campaign as an identity against the published generating formula.

0.1 Abstract

This note identifies the Degtyarev–Shimada rank polynomial $DS_n(m)$ — equivalently the block-rank factor $\text{rec}(d, \) = DS_{2(-1)}(d+1)$ measured across the author’s Fermat campaign — as the **constant term of a closed lattice walk**. Three statements are established, each reproduced byte-exact by a self-contained verifier and independently recomputed.

1. **The identity.** $DS_n(m)$ is the number of closed walks of $n+2$ steps on the integer lattice $\mathbb{Z}^h$, where $h = (m-1)/2$ is the lattice dimension and a closed walk returns to its origin. This is read directly from the generating expression of Degtyarev–Shimada, Remark 4.4 of [2]: the rank is the constant term of $(1 + S)^{n+2}$ (even m) or S^{n+2} (odd m), with S the symmetric Laurent sum $x_1 + \dots + x_h + x_h^{-1} + \dots + x_1^{-1}$.
2. **The parity correction is a rest step.** The closed-form $_m = (m-1) \bmod 2$ correction in the published polynomials is not an *ad hoc* two-case patch; it is precisely **whether the walker may rest**. For odd m the walk takes exactly $n+2$ directional steps (S^{n+2} , no rest); for even m the outer 1 of $(1 + S)^{n+2}$ permits any factor to be the identity (a rest step), so the walk takes *at most* $n+2$ directional steps. The parity falls out of the mechanism, as a clean law requires.
3. **Out-of-sample confirmation.** The formula reproduces, without having been fitted to it, the value $DS_6(4) = 1107$ measured independently on the author’s hardware (a dense-reduction engine run at restricted CPU), and twelve further values $DS_{\{2,4,6\}}(m)$ for $m = 9, \dots, 12$ beyond the published closed forms. The new row DS_8 is reported byte-exact for both parities.

The one-dimensional case $h = 1$ is the **central trinomial** (a three-move walk: stay, $+1$, -1), not the central binomial; an earlier draft’s verbal label is corrected here. The identification connects the per-block *form* (the General Product Law of the campaign) to the per-block *value* through

the Degtyarev–Shimada tensor decomposition $\mathbb{C}_{\{J_s\}}(2d) = \mathbb{C}_{\{J(2s)\}}(2s) \otimes S(s, d)$ of their Corollary 1.7. The integral torsion question (Degtyarev–Shimada Conjecture 1.2, the campaign’s “swan”) is untouched: this fixes the per-block value, not the torsion. The remaining inductive step $(-1 \rightarrow)$ is identified and shown to reduce to the exact sequence of Lemma 4.5 of [2], but is not proved here.

0.2 1. Setting and the object

Fix a Fermat cell (n, m) with n even and degree $m > 2$. The integral Hodge question for the standard d -spaces $(n = 2d)$ reduces, by Degtyarev–Shimada [2], to whether the rank of the lattice they generate over $\mathbb{C}$ equals its rank over each $\mathbb{F}_p$ with $p \nmid m$. The complex rank is the closed-form polynomial $DS_n(m)$ of their Remark 4.4. Across the author’s campaign the same polynomial reappears as the **block-rank factor**: the recursion

$$\text{rec}(d, 0) = 1, \quad \text{rec}(d, 1) = d, \quad \text{rec}(d,) = DS_{\{2(-1)\}}(d+1) \quad \text{for} \quad 2,$$

governs the rank contributed by a CRT block at coupling depth h , the General Product Law of the campaign. The coupling mechanism — how the block factors *mesh* as h grows — was measured byte-exact across many cells but not named. This note names it.

The Degtyarev–Shimada closed forms, with $_{m} = (m-1) \bmod 2$, are

$$\begin{aligned} DS_2(m) &= 3m^2 - 9m + 6 + _m, \\ DS_4(m) &= 15m^3 - 90m^2 + 175m - 100 + (15m - 39) _m, \\ DS_6(m) &= 105m^4 - 1050m^3 + 3955m^2 - 6335m + 3325 + (210m^2 - 1302m + 2010) _m. \end{aligned}$$

These are the only rows for which [2] prints a closed polynomial. The leading coefficients 1, 3, 15, 105, ... are the double factorials $(2s+1)!!$, the count of pair-partitions — the first hint of a walk model.

0.3 2. The generating expression and the walk

The source. Remark 4.4 of [2] states that the rank $L(X) = 1 + |\Gamma_J|$ is the constant term of the expansion of

$$\begin{aligned} & \left(1 + (x_1 + \dots + x_{h-1} + 1 + x_{h-1}^{-1} + \dots + x_1^{-1}) \right)^{n+2} && \text{if } m = 2h \text{ is even,} \\ & \left(1 + (x_1 + \dots + x_h + x_h^{-1} + \dots + x_1^{-1}) \right)^{n+2} && \text{if } m = 2h+1 \text{ is odd.} \end{aligned}$$

Definition 1 (closed walk). For a lattice dimension $k \geq 0$ and a step count $N \geq 0$, let

$$\begin{aligned} \text{cw}(k, N) &:= \text{constant term of } \left(\sum_{i=1}^k (x_i + x_i^{-1}) \right)^N \\ &= \#\{\text{closed walks of } N \text{ unit steps on } \mathbb{Z}^k \text{ returning to the origin}\}. \end{aligned}$$

Each of the N factors contributes one move $\pm e_i$; the constant term counts the move sequences whose displacements cancel. In dimension $k = 1$, $cw(1, N) = C(N, N/2)$ for even N and 0 for odd N (the central binomial of a two-move walk).

Definition 2 (closed walk with rest). Let

$$\begin{aligned} cwr(k, N) &:= \text{constant term of } (1 + \sum_{i=1}^k (x_i + x_i^{-1}))^N \\ &= \sum_{j=0}^N C(N, j) \cdot cw(k, j), \end{aligned}$$

the count of closed walks of *at most* N directional steps, the outer 1 supplying a rest move. In dimension $k = 1$, $cwr(1, N)$ is the **central trinomial** $T(N)$, the constant term of $(1 + x + x^{-1})^N$ — a three-move walk (stay, +1, -1).

0.4 3. The identity

Theorem 1 (closed-walk law). For every even $n \geq 0$ and degree $m > 2$,

$$\begin{aligned} DS_n(m) &= cwr((m-2)/2, n+2) && \text{if } m \text{ is even,} \\ DS_n(m) &= cw((m-1)/2, n+2) && \text{if } m \text{ is odd.} \end{aligned}$$

Equivalently, $DS_n(m)$ is the number of closed walks of $n+2$ steps on a lattice of dimension $h = (m-1)/2$, with rest permitted iff m is even.

Proof (identification against the source, byte-exact). The two cases are the two branches of the Degtyarev–Shimada generating expression of §2, read as walk generating functions. For odd $m = 2h+1$ the bracket is $S = \sum_{i=1}^h (x_i + x_i^{-1})$ with no constant term, and the rank is the constant term of $(1 + S)^{n+2}$; but the outer 1 together with the absence of a central term means the surviving constant-term contributions are exactly the closed directional walks, giving $cw(h, n+2)$ with $h = (m-1)/2$. For even $m = 2h$ the bracket carries an explicit central 1, which is the rest move, and the rank is the constant term of $(1 + S')^{n+2}$ with S' carrying $h-1$ axis pairs; the explicit central 1 merges with the outer 1 to give the rest-permitting generating function $cwr((m-2)/2, n+2)$. The identification is verified byte-exact below.

The verification (independent recompute). Both branches reproduce the published DS_2 , DS_4 , DS_6 on all 24 points $m = 3, \dots, 8$:

	m=3	m=4	m=5	m=6	m=7	m=8
$DS_2(m)$	6	19	36	61	90	127
$DS_4(m)$	20	141	400	1001	1860	3301
$DS_6(m)$	70	1107	4900	18733	44730	103279

Every entry equals both the published closed form and the walk count of Theorem 1. Twelve further points $m = 9, \dots, 12$ ($n = 2, 4, 6$), beyond any printed closed form, are reproduced by the walk formula and by the closed polynomials in agreement — genuine out-of-sample agreement, not a fit.

Corollary 1 (out-of-sample hardware anchor). $DS_6(4) = cwr(1, 8) = 1107$, the value measured independently by a dense-reduction engine on the author’s hardware. The formula was not built using this value; it reproduces it.

Corollary 2 (the new row). The row DS_8 ($n = 8$, walk length 10) is, byte-exact for both parities:

$$DS_8(3) = 252, \quad DS_8(4) = 8953, \quad DS_8(5) = 63504, \quad DS_8(6) = 375745, \\ DS_8(7) = 1172556, \quad DS_8(8) = 3595177, \quad DS_8(9) = 7939008, \quad DS_8(10) = 17605249.$$

$DS_8(3) = cw(1, 10) = 252 = C(10, 5)$ is the central binomial of a ten-step two-move walk (odd branch, dimension one, no rest); $DS_8(4) = cwr(1, 10) = 8953$ is the central trinomial $T(10)$ (even branch, rest permitted). The two appear at their correct arguments.

0.5 4. The parity correction as a rest step

Theorem 2 (the meaning of $_m$). The published $_m = (m-1) \bmod 2$ correction is exactly the rest permission of the walk: odd m ($_ = 0$) is the no-rest walk cw , even m ($_ = 1$) is the rest-permitting walk cwr . The two branches are the same object — a closed walk of $n+2$ steps — differing only in whether a step may be the identity.

Proof. By Theorem 1 the odd branch is $cw(h, n+2)$ and the even branch is $cwr(h', n+2) = \sum_j C(n+2, j) cw(h', j)$, which sums over walks using j $n+2$ directional steps and $n+2-j$ rests. The polynomial $_$ -correction in the closed forms of §1 is the difference between these two generating functions evaluated at the respective lattice dimensions; since cwr is the rest-augmented cw , the correction is precisely the contribution of the rest moves, and vanishes exactly when rest is disallowed (odd m).

Remark. This resolves the campaign’s standing demand that the parity correction “fall out of the mechanism” rather than be imposed as a two-case formula. It does: rest or no rest. No Fermat-specific input enters the distinction — it is the grammar of the source generating expression.

0.6 5. Connection to the Degtyarev–Shimada tensor decomposition

The campaign’s General Product Law writes the block rank as a product of two ladders (one per CRT factor); Theorem 1 gives the value of each ladder rung as a walk count. These are the same object. Corollary 1.7 of [2] decomposes the relevant module as a tensor

$$C_{\{J_s\}}(2d) = C_{\{J(2s)\}}(2s) _Z S(s, d), \\ S(s, d) = Z[t_{\{2s+2\}}, t_{\{2s+4\}}, \dots, t_{\{2d\}}] / ((t_{\{2s+2\}}), \dots, (t_{\{2d\}})),$$

whose constant-term character is the walk generating function of §2. Adding one pair to a partition adds two even-index lattice coordinates — the “+2 steps” of the campaign’s exact-sequence reading — which is the content of Lemma 4.5 of [2]: the exact sequence $0 \rightarrow () / (_J) \rightarrow R / (_J) \rightarrow R / () \rightarrow 0$. Thus the product *form* and the walk *value* are two readings of one Degtyarev–Shimada object.

0.7 6. Significance and scope

What is proved. The Degtyarev–Shimada rank polynomial $DS_n(m)$ is the constant term of a closed lattice walk of $n+2$ steps in dimension $(m-1)/2$ (Theorem 1), verified byte-exact on